(a) Find $\int x^2 \cos 2x \, \mathrm{d}x$

(4)

(b) Hence solve the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}t} = \left(\frac{t \cos t}{y} \right)^2$$

giving your answer in the form $y^n = \mathrm{f}(t)$ where n is an integer.

(5)